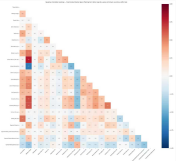
Customer Churn Prediction & Retention Analysis

1. Executive Summary

Core Objective
Define target (Churn) and remove leakage variables. Drop identifiers and redundant features (e.g., Total Charges, CLTV). CLTV validated via correlation with existing variables.

Preprocessing:
- Numerical: median imputation + standardization
- Categorical: mode imputation + one-hot encoding

2. Data Processing



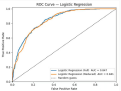
(a) Correlation Heatmap

(b) VIF

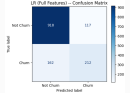
Low–moderate correlations indicate weak direct relationships.
VIF reveals multicollinearity from one-hot encoding and overlapping features.

3. Model Performance

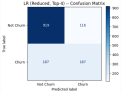
3.1 Logistic Regression



(a) ROC Curve



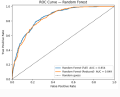
(b) Full Features



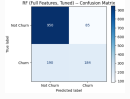
(c) 4 Features

The ROC curves show that the full model achieves a slightly higher AUC (0.847 vs 0.841), indicating better discrimination ability. The confusion matrices also suggest that the full model identifies more churn cases, while the reduced model misses more positives.

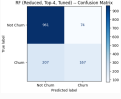
3.2 Random Forest



(a) ROC Curve



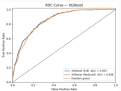
(b) Full Features



(c) 4 Features

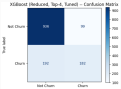
Full model: Acc 0.805, Rec 0.492, F1 0.572, AUC 0.854
Reduced model: Acc 0.801, Rec 0.447, F1 0.543, AUC 0.849
Full model detects more churn (184 vs 167), with fewer misses.

3.3 XGBoost



(a) ROC Curve

(b) Full Features



(c) 4 Features

Using the optimized parameters, we train two models on different feature sets. With the full feature set, the model achieves ROC-AUC of 0.855. With the reduced feature set, performance

4. Final Conclusion

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression (Full)	0.8020	0.6444	**0.5988**	**0.6031**	0.8468
Logistic Regression (Reduced)	0.7850	0.6172	0.5000	0.5524	0.8415
Random Forest (Full)	0.8048	**0.6840**	0.4920	0.5723	0.8543
Random Forest (Reduced)	0.8006	0.6929	0.4465	0.5431	0.8490
XGBoost (Full)	**0.8070**	0.6723	0.5321	0.5940	**0.8549**
XGBoost (Reduced)	0.7935	0.6477	0.4866	0.5557	0.8479

Among all six experiments, XGBoost with the full feature set achieves the highest ROC-AUC (0.8549), indicating the best overall discriminative ability.

Compared with Logistic Regression and Random Forest, XGBoost consistently delivers stronger performance, especially in capturing non-linear relationships and improving recall for churn customers.

Additionally, the full feature set is preferred, as it preserves all available information and allows the model to leverage complex interactions among variables, which is critical for accurately predicting customer churn.

Therefore, XGBoost (Full Features) is selected as the final model, as it provides the best balance of predictive performance and comprehensive feature utilization.